

Electrochemistry

Subject / Theme :

Date: / /

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- Electrolytes and Non-electrolytes.
- Strong electrolytes and weak electrolytes.
- Ionization and dissociation
- Mobility of Ions.
- Arrhenius theory of Ionization.
- Faraday's Law of electrolytes.
- Transport number.
- Exhibition of electrolysis process in a cell.
- Reversible cell and Irreversible cell and molar conductance
- Mathematical Problem.

Electrolytes:-

An electrolyte is a compound that conducts electric current when it is in aqueous solution state or in melted state.

Ex - NaCl , CaSO_4 etc.

Characteristics:-

- ① All Ionic compounds are electrolytes.
- ② When Ionic compounds dissolve, they break apart into ions which are then able to conduct current.
- ③ They form ions when they are placed in water.
- ④ Insoluble ionic compound such as CaCO_3 are electrolyte.

Non-electrolytes:-

An non-electrolyte is a compound that doesn't conduct electric current in either aqueous solution or in melted state.

Ex - CH_4 , NH_3 , Sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) etc.

Characteristics:-

- ① Almost all covalent compounds are ~~are~~ electrolytes or non-electrolytes.
- ② When these compounds dissolve in water, they don't produce ions.
- ③ They don't form ions when they are placed in water.

Difference between electrolytes and non-electrolytes:-

Electrolytes	Non-electrolytes.
1. Substances when they are dissolved in water allow for conducting electricity are called electrolytes.	1. Substances when they are dissolved in water doesn't allow for conducting electricity are called non-electrolytes.
2. They produce ions in water.	2. They don't produce ions in water.
3. These substances are formed by ionic bond.	3. These substances are formed by covalent bond.
4. Acid, bases and salts are electrolytes.	4. Sugar, honey etc are covalent compounds or non-electrolytes.

Strong Electrolytes:

A strong electrolyte is a solution that completely or almost completely ionizes or dissociates in a solution.

→ Strong electrolytes are good conductors of electricity.

→ The strong electrolytes may be strong acids, strong bases or soluble ionic salts.

→ The strong electrolytes are:-

i. Strong Acids:- HCl , H_2SO_4 , HNO_3 , HClO_4 , HBr , HI etc.

ii. Strong Bases:- NaOH , KOH , Ca(OH)_2 , Mg(OH)_2 etc.

iii. Salts:- All salts are strong electrolytes.

NaCl , KCl , CaSO_4 , $\text{Ca(NO}_3)_2$ etc.

Weak Electrolytes:-

A weak electrolyte is a solution that isn't completely ionized or dissociates in a solution.

→ The weak electrolytes may be organic acids, organic bases or few salts that aren't soluble in water.

→ The weak electrolytes are:-

i. Organic Acids:- CH_3COOH (acetic acid)

HOOC-COOH (oxalic acid)

H_2SO_3 (sulphurous acid) etc.

ii. Organic Bases:- $\text{C}_2\text{H}_5\text{NH}_2$ (alkyl amine).

iii. A few salts like Mercury II chloride (HgCl_2), Lead II acetate ($\text{Pb(CH}_3\text{COOH)}_2$).

Roles of electrolytes in our body:-

1. Regulate blood and muscle function.
2. Balance the amount of water in the body.
3. Move water from our body.
4. Keep balancing body's pH level.
5. Help to rebuild damaged tissues.
6. Move nutrients into cell.
7. Keep blood pressure stable.
8. Keep heart rate and rhythm steady.

Ionization:-

Ionization refers to the reaction in which the polar covalent compounds are converted into ions in water.

→ It is the process that involves the formation of ions.

→ It involves the creation of charges across the participating species.

Dissociation:-

Dissociation refers to the separation of ions which are already present in electrovalent or ionic compounds.

→ It is the process of breaking up of a moiety (Half) into its constituent atoms, molecules and ions.

→ It occurs due to a weak bond between species.

☐ Difference between ionization and dissociation:-

Ionization	Dissociation
1. It is the process to create or produces new charged particles.	1. It is the separation of charged particles which already exists in a compound.
2. It involves polar covalent compounds or metals.	2. It involves ionic compounds.
3. It is Irreversible.	3. It is reversible.
4. Involve covalent bond between atoms.	4. Involve ionic bond in compound.
5. Always produces charged particles.	5. It produces charged particles or electrically neutral particles.

☐ Conductance:-

Conductance is the ability of an element to conduct electric current.

Conductance is expressed by G .

So, $G = \frac{1}{R}$.

Unit of conductance is Siemens (S).

☐ Molar Conductance:-

The ability of conducting power of total power of all the ions produced by 1 mole of an electrolyte in the given solution.

It is represented by χ

$$\chi = K \times V$$

$$\chi = K \times 1000 / M$$

Where,

M = Molarity of the solⁿ (M)

V = Volume of the solⁿ (c.c)

K = Specific conduction of the solⁿ.

☐ Ionic Mobility:-

Ionic mobility is defined as the distance travelled by an ion per second under potential gradient of ~~one~~ one volt per meter.

☐ Potential Gradient:-

Potential Gradient is given by the potential difference applied at the electrodes divided by distance between the electrodes.

$$\text{Ionic Mobility} = \frac{\text{Speed of the ions in ms}^{-1}}{\text{Potential Gradient (Vm}^{-1}\text{)}}$$

→ Unit of ionic mobility: $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$.

Transport Number:-

The fraction of total electric current carried by the cation or the anion is termed as its transport number or Hittorf's number.

If v_+ and v_- be the speed of the cation and anion respectively.

$$\begin{aligned} \text{Transport Number of the cation } T_+ &= \frac{v_+}{v_+ + v_-} \\ \text{" " " " anion } T_- &= \frac{v_-}{v_+ + v_-} \end{aligned}$$

Ration of the transport number T_+ and T_- ,

$$r = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+} \quad [T_+ + T_- = 1]$$

$$t_- = \frac{1}{1 + r}$$

1. The speed ratio of silver and nitrate ions in a solution of silver nitrate electrolysed between silver electrodes is 0.916. Calculate the transport numbers of Ag^+ and NO_3^- .

Given,

$$r = 0.916$$

We know,

$$t_{\text{NO}_3^-} = \frac{1}{1+r}$$

$$= \frac{1}{1+0.916}$$

$$= 0.521$$

$$\therefore t_{\text{Ag}^+} = 1 - t_{\text{NO}_3^-}$$

$$= 1 - 0.521$$

$$= 0.479$$

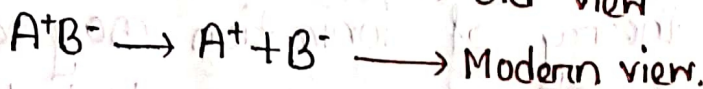
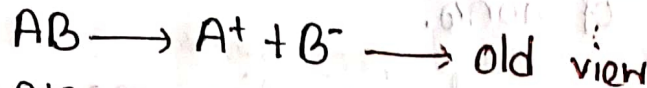
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1. Arrhenius Law $\rightarrow$ Natural molecule

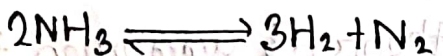
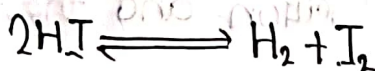
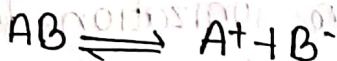
2. Faraday's law of electrolysis.

3. Electrolysis Process in a cell.



Arrhenius law.

Sum of the charge = 0



$$K_c = \frac{[H_2][I_2]}{[HI]^2}$$

According to the law of mass action:

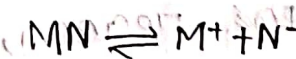
$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

 $\rightarrow$ Equilibrium constant.

সাম্য স্থিতি

Dissociation Constant $\rightarrow K_a$ বিয়োজন স্থিতি.

$$K_a = \frac{[M^+][N^-]}{[MN]}$$



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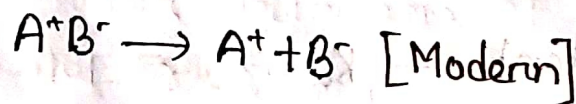
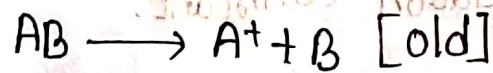
Arrhenius Theory of Ionization:-

Savante Arrhenius theory studied the conduction of current through water solution of electrolytes. He came to believe that the conductivity of solutions was due to the presence of ions.

Arrhenius theory of ionization may be stated as:-

- ① When dissolved in water, neutral electrolyte molecules are split up into two types of charged particles. These particles are called ions and the process was termed as ionization. The positively charged are called cation and the negatively charged are called anion.

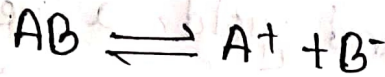
In this modern form, the theory assumes that the ions are already present in the solid electrolytes and are held together by electrostatic force. When placed in water, these neutral molecules dissociate to form separate cations and anions.



For these reason, this theory may be referred to as theory of electrolytic dissociation.

② The ions present in solution constantly reunite to form neutral molecules.

Thus there is a state of equilibrium between the dissociated molecules and ions.



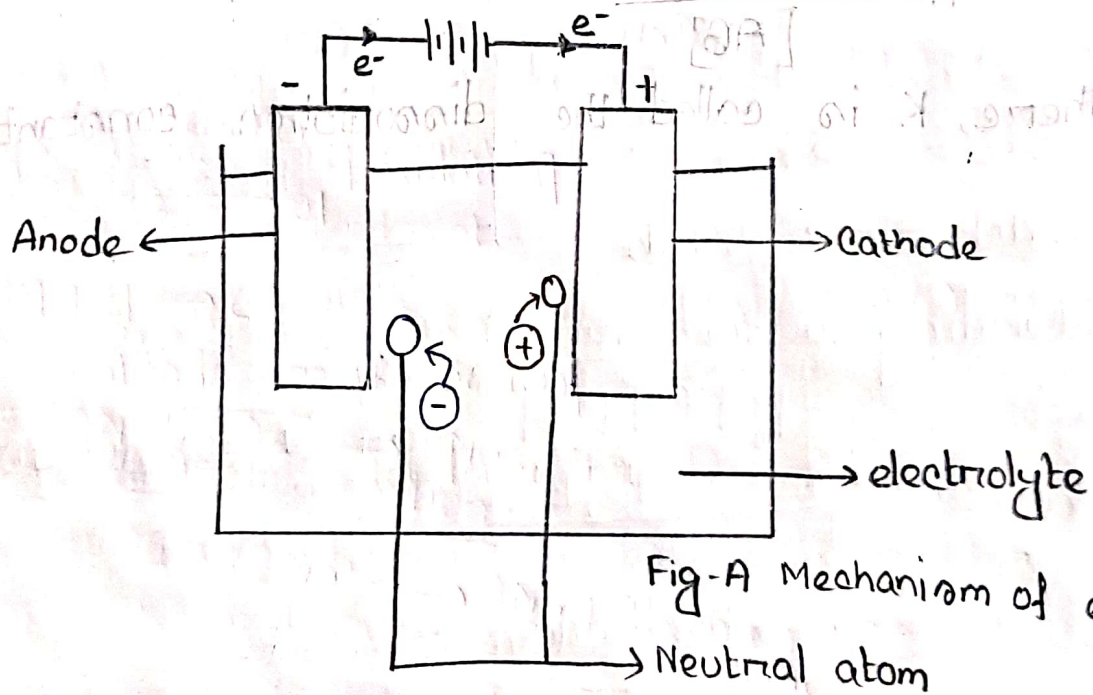
Applying the law of mass Action to the ionic equilibrium, we have

$$K = \frac{[A^+][B^-]}{[AB]}$$

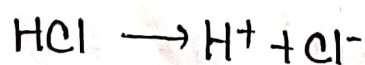
where, K is called the dissociation constant.

Electrolysis process in a cell:-

The method of electrolysis is illustrated in Fig-A. The cation migrate to the cathode and form a neutral atom by accepting electrons from it. The anions migrate to the anode and yield a neutral particle by transference of electrons by anions and gain of electrons by cations at their respective electrode. Chemical reaction takes place.



Let us consider electrolysis of HCl as an example. In solution, HCl is ionised.



Anode ऽऒ ऒऒऒऒ
cathode ऽ ऒऒऒऒऒ

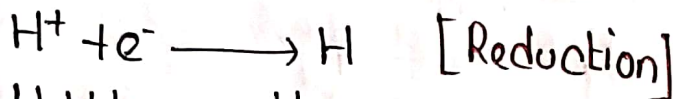
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In this electrolysis, Cl^- ions will move towards the anode and H^+ ions will move towards the cathode. At the electrode, the following reaction will take place,

At cathode:-

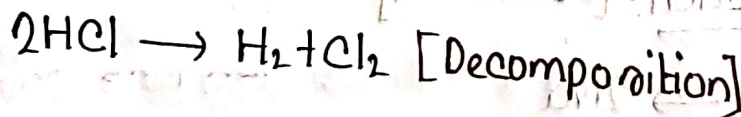


At anode:-



Pairs of H atom combines to form H_2 molecule and pairs of Cl atom combines to form Cl_2 molecule.

Combined Reaction:-

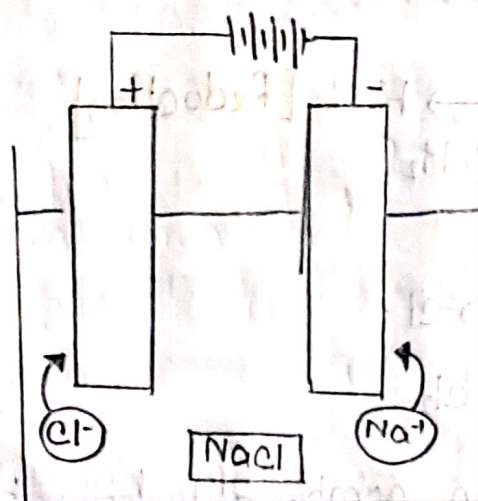
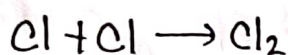
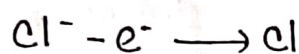
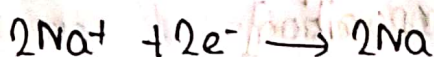
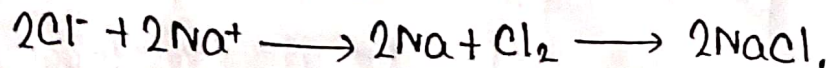


2nd Methode:-

class

The phenomenon of decomposition of electrolytes in a cell by passing current through its solution is termed electrolysis.

Ar

Anode Reaction:- [Oxidation]Cathode reaction:- [Reduction]combined reaction:-

Mobility of Ions / Relative speed of ions:-

It is already considered that ions move to the oppositely charged electrodes under the influence of the electric current. But the speeds of cations migrating towards the cathode and the speed of anions migrating towards the anode are not necessarily the same. However the speed of a cation moving away from the anode will be proportional to the fall of concentration of cations around the anode.

Similarly, the speed of anions moving away from the cathode will be proportional to the fall of concentration of anions around the cathode.

Hittorf's gave a general rule is known as Hittorf's Rule:-

"The loss of concentration around any electrode is proportional to the speed of ions moving away from it."

Illustration of Hittorf's Rule:-

In the figure,

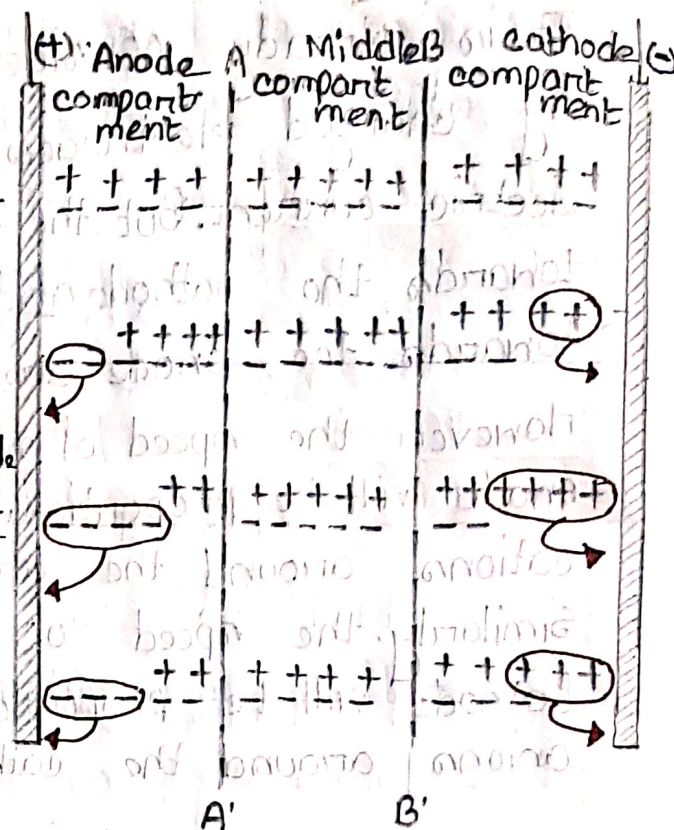
A is the anode
and C is the cathode.

AA' and BB' are two
imaginary planes which divide
the cell into three
compartments.

→ Anode compartment.

→ Cathode compartment.

→ Middle compartment.



The sign (+) represents a cation and (-) represents an anion.

Before electrolysis, let there be 13 ions-pairs in the cell.

The number of ions-pairs in the two outer compartments is 4 each and there are 5 ion-pairs in the middle compartment (position I).

i.) Let the anions alone be capable of movement:-

When 2 anions have moved towards the anode, we get the position (II). The cations have not moved at all. The number of discharged anions and cations is the same viz. 2.

The concentration in the anode compartment has not altered while in the cathode compartment it has fallen by two ion-pairs.

ii.) Let the anions and cations move at the same rate:-

When 2 ions of each type have crossed towards the opposite electrodes, we get the condition shown in (III).

The number of discharged anions and cations is the same viz 4.

Anode and cathode compartment has fallen to the same extent by 2 pairs.

iii.) Let the cations move at twice the speed of the anions:-

When cation have moved to the cathode compartment, one anion passes into the anode compartment. This state is shown in (IV). Total number of discharged anions and cations is again same. In the concentration, Cathode compartment has fallen by one ion-pairs, Anode compartment has fallen by two ion-pairs.

1. A solution of silver nitrate containing 12.14g of silver in 50ml of solution was electrolysed between platinum electrodes. After electrolysis, 50ml of the anode solⁿ was found to contain 11.55g of silver, while 1.25g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

Sol:-

Weight of Ag in 50ml of the solution before electrolysis = 12.14g.

Weight of Ag in 50ml of the solution after electrolysis = 11.55g.

Fall in concentration of Ag = $(12.14 - 11.55) \text{ g}$

$$= 0.59 \text{ g}$$

$$= \frac{0.59}{108}$$

$$= 0.0055 \text{ g eq.}$$

Weight of Ag deposited in silver coulometer 1.25 g

$$= \frac{1.25}{108}$$

$$= 0.0116 \text{ g eq.}$$

Hence,

Transport number of Ag^+ (t_{Ag^+})

$$= \frac{\text{fall in conc around anode}}{\text{No of g eqvt deposited in silver coulometer}}$$

$$= \frac{0.0055}{0.0116}$$

$$= 0.474$$

 $\therefore$ Transport Number of NO_3^- ($t_{\text{NO}_3^-}$) $= 1 - 0.474$

$$= 0.526$$

(Ans)

2. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91g respectively. Calculate the transport number of Cu^{2+} and SO_4^{2-} ions.

Sol:-

Weight of copper in the anode liquid before electrolysis-

0.79g.

Weight of copper in the anode liquid after electrolysis

0.91g.

$$\therefore \text{Increase in weight} = 0.91 - 0.79 \text{ g} \\ = 0.12 \text{ g.}$$

$$\text{Increase in weight of copper cathode in coulometer} = \cancel{\phi} \frac{0.153}{0.153 \text{ g [Given]}}$$

But actual increase = 0.12g.

Fall ϕ in concentration due to migration of Cu^{2+} ions

$$= 0.153 - 0.12$$

$$= 0.033 \text{ g.}$$

$$\therefore \text{Transport Number of } \text{Cu}^{2+} \text{ ion } (T_{\text{Cu}^{2+}}) = \frac{0.033}{0.153}$$

$$= 0.215$$

$$\text{Transport Number of } \text{SO}_4^{2-} (T_{\text{SO}_4^{2-}}) = 1 - 0.215$$

$$= 0.785$$

(Ans)

3. The speed ratio of silver and nitrate ions in a solution of silver nitrate electrolysed between silver electrodes is 0.916. Find the transport number of the two ion.

Sol:-

Given,

$$r = 0.916$$

$$\therefore T_{\text{NO}_3^-} = \frac{1}{1+r} = \frac{1}{1+0.916} = 0.521$$

$$\therefore T_{\text{Ag}^+} = 1 - 0.521$$

$$= 0.479$$

(Ans)

Faraday's Law of Electrolysis:-

First law of Faraday:-

"The amount of substances deposited or liberated at an electrode during electrolysis is directly proportional to the quantity of charge (electricity) passed through the electrolyte solution."

If m is the mass of the substance (in grams) deposited on electrode by passing Q coulombs of electricity, then,

$$m \propto Q$$

$$m \propto It \quad [Q = It]$$

$$\boxed{m = ZIt}$$

where, Z is the constant known as the electrochemical equivalent of the substance (electrolyte) and I is the strength of current in amperes and t is the time in seconds for which the current has been passed.

If $I = 1 \text{ A}$, $t = 1 \text{ s}$,

then $m = Z$.

Thus electrochemical equivalent is the amount of a substance deposited by 1 ampere current passing for 1 second.

$$Z = \frac{\text{Equivalent mass of substance}}{\text{Faraday constant}} = \frac{E}{F} = \frac{E}{96500}$$

Faraday constant,

$$F = 96500.$$

Faraday's 2nd Law of Electrolysis:-

The mass of a substance deposited or liberated at any electrode on passing a certain amount of charge is directly proportional to its chemical equivalent weight

i.e.,

$$m \propto E$$

$$\boxed{\frac{m_1}{m_2} = \frac{E_1}{E_2}}$$